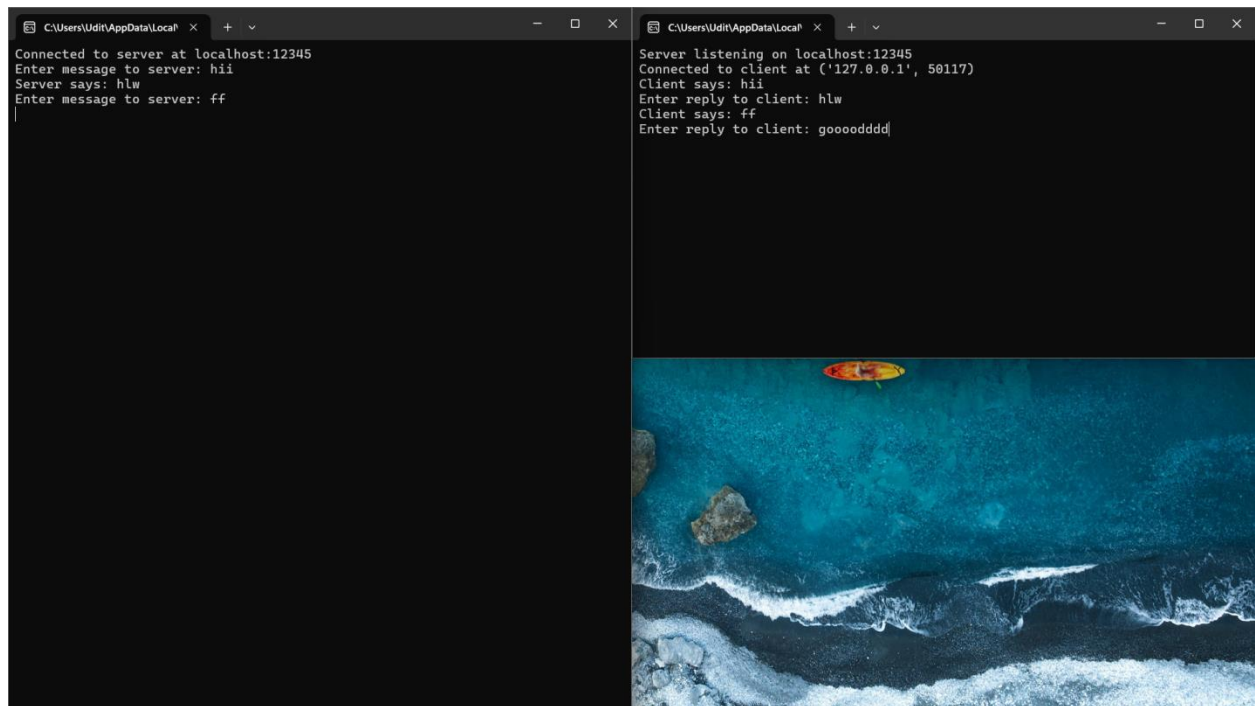


CAPSTONE-4

Name-Satyam Parida

Regd_no-2141016403

PYTHON LOCAL CODE -



The image shows two side-by-side terminal windows. The left window displays the output of a client program connected to a server at localhost:12345. The right window displays the output of a server program listening on localhost:12345. The server window also includes a decorative image of a beach and ocean at the bottom.

```
C:\Users\Udit\AppData\Local>
Connected to server at localhost:12345
Enter message to server: hii
Server says: hlw
Enter message to server: ff

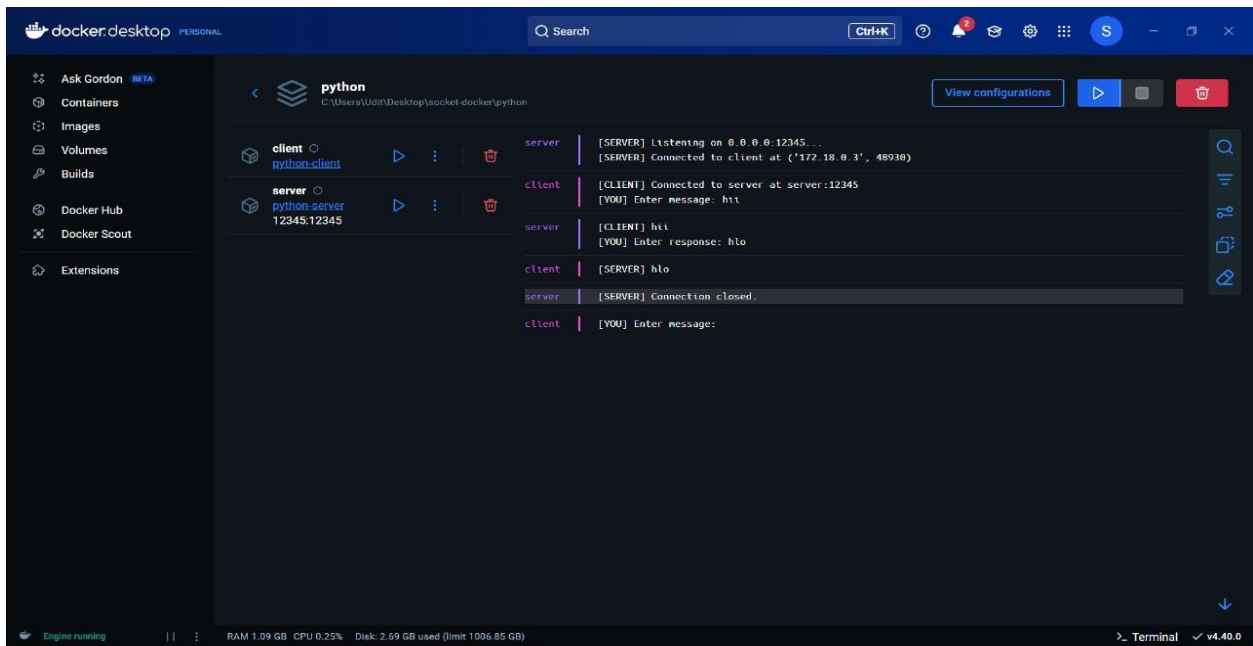
C:\Users\Udit\AppData\Local>
Server listening on localhost:12345
Connected to client at ('127.0.0.1', 50117)
Client says: hii
Enter reply to client: hlw
Client says: ff
Enter reply to client: gooooodddd
```



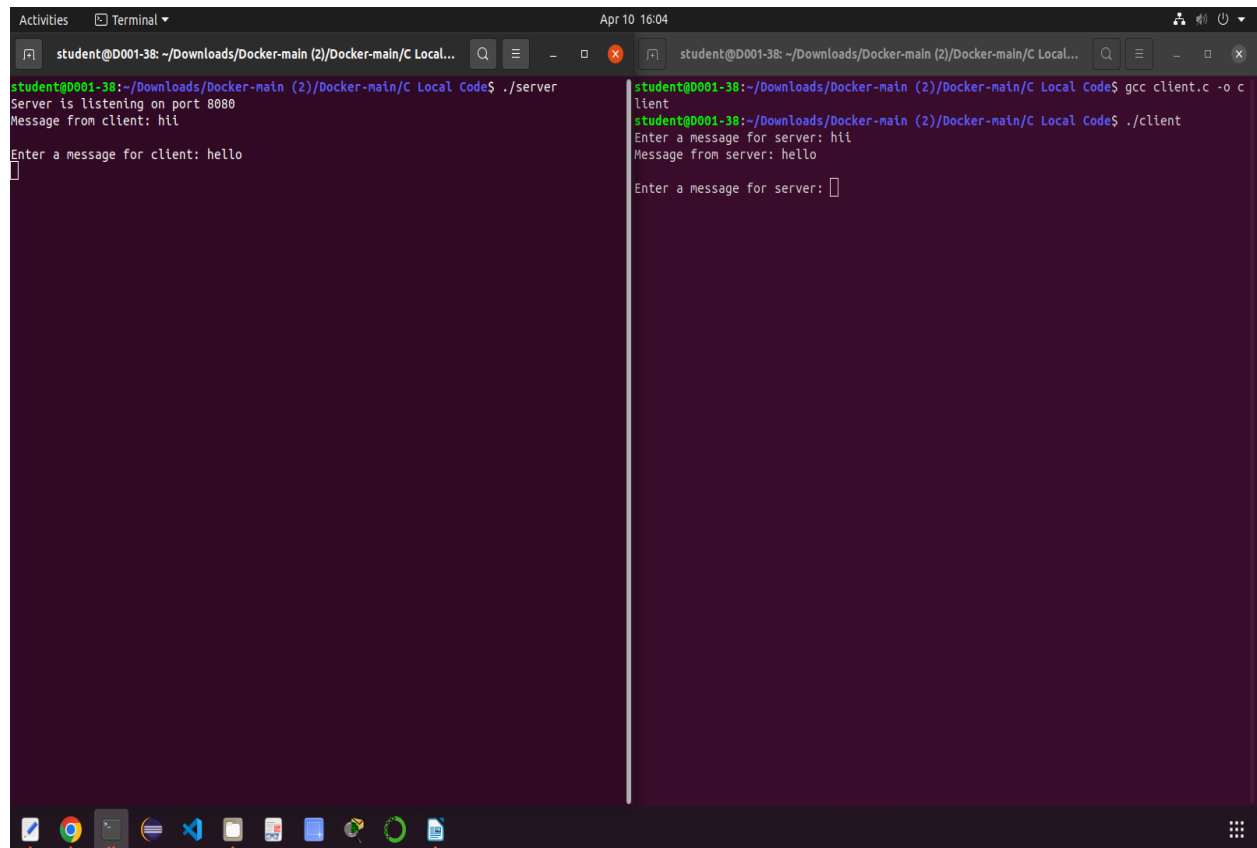
PYTHON DOCKER CODE –

```
Command Prompt - docker-c x Windows PowerShell x + v
=> [client internal] load build context
=> => transferring context: 318 0.0s
=> CACHED [client 3/3] COPY client.py . 0.0s
=> [client] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:440d246e66c8a449414b339a5deee208e0eebce54d235e691fb35aea208c236c 0.0s
=> => naming to docker.io/library/python-client 0.0s
=> [client] resolving provenance for metadata file 0.0s
[+] Running 4/4
✓client Built 0.0s
✓server Built 0.0s
✓Container python_socket_server Running 0.0s
✓Container python_socket_client Running 0.0s
Attaching to python_socket_client, python_socket_server
python_socket_client | hii
python_socket_server | [CLIENT] hii
python_socket_server | [YOU] Enter response: hlo
python_socket_client | [SERVER] hlo

View in Docker Desktop View Config Enable Watch
```



C LOCAL CODE -



The image shows two terminal windows side-by-side. The left window shows the execution of a server program. The right window shows the compilation and execution of a client program.

```
student@D001-38: ~/Downloads/Docker-main (2)/Docker-main/C Local Code$ ./server
Server is listening on port 8080
Message from client: hii
Enter a message for client: hello

```

```
student@D001-38:~/Downloads/Docker-main (2)/Docker-main/C Local Code$ gcc client.c -o client
student@D001-38:~/Downloads/Docker-main (2)/Docker-main/C Local Code$ ./client
Enter a message for server: hii
Message from server: hello
Enter a message for server:

```

C DOCKER CODE –

```
Windows PowerShell
0.1s
=> => writing image sha256:c3fb53dfbdee34a326617f896ad72a5d63f2c8cfc465b4b6
73760fa66201baf6 0.0s
=> => naming to docker.io/library/cdockerfile-server
0.0s
=> [server] resolving provenance for metadata file
0.0s
=> [client internal] load build definition from Dockerfile
0.0s
=> => transferring dockerfile: 291B
0.0s
=> [client internal] load .dockerignore
0.0s
=> => transferring context: 2B
0.0s
=> [client internal] load build context
0.0s
=> => transferring context: 58B
0.0s
=> CACHED [client 3/6] COPY server.c .
0.0s
=> CACHED [client 4/6] COPY client.c .
0.0s
=> CACHED [client 5/6] RUN gcc -o server server.c
0.0s
=> CACHED [client 6/6] RUN gcc -o client client.c
0.0s
=> [client] exporting to image
0.0s
=> => exporting layers
0.0s
=> => writing image sha256:b14a72e6db0331b26447d7e6a88c74abf5ce6b8408d3957e
acb95631e736bc49 0.0s
=> => naming to docker.io/library/cdockerfile-client
0.0s
=> [client] resolving provenance for metadata file
0.0s
[+] Building 2/2
✓client Built 0.0s
✓server Built 0.0s

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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Udit\Desktop\socket-docker\C\C Docker File> docker-compose run c
lient
time="2025-04-10T16:13:31+05:30" level=warning msg="C:\\Users\\Udit\\Desktop
\\socket-docker\\C\\C Docker File\\docker-compose.yml: the attribute 'versio
n' is obsolete, it will be ignored, please remove it to avoid potential conf
usion"
[+] Creating 1/1
  ✓Container chat_server Created 0.1s
[+] Running 1/1
  ✓Container chat_server Started 0.3s
Connected to server!
Client: hello

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Udit\Desktop\socket-docker\C\C Docker File> docker-compose run s
erver
time="2025-04-10T16:13:14+05:30" level=warning msg="C:\\Users\\Udit\\Desktop
\\socket-docker\\C\\C Docker File\\docker-compose.yml: the attribute 'versio
n' is obsolete, it will be ignored, please remove it to avoid potential conf
usion"
[+] Creating 1/1
  ✓Network cdockerfile_chatnet Created 0.1s
Server waiting for connection...
```

